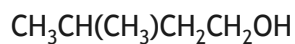


5. Fusel oil is formed as a by-product during the production of bioethanol for fuel. It is a mixture of several alcohols.

- (a) The shortened structural formula of one of the alcohols contained in fusel oil is shown.



State the name of this alcohol.

1

- (b) Propan-1-ol is also found in fusel oil.

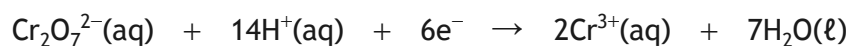
Propan-1-ol is reacted with an oxidising agent to produce propanal.

- (i) Complete the ion-electron equation for the oxidation reaction.

1



- (ii) Acidified potassium dichromate can be used as the oxidising agent and reacts as shown below.



Suggest why the potassium dichromate must be acidified.

1

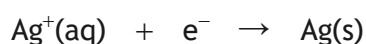


5. (b) (continued)

- (iii) State the colour change that would be observed when propan-1-ol reacts with acidified potassium dichromate.

1

- (iv) The equation for the reduction of another oxidising agent that could be used to oxidise propan-1-ol is shown below.



Name the reagent that provides this oxidising agent.

1

- (v) State why 2-methylbutan-2-ol cannot be oxidised using these oxidising agents.

1

- (vi) In the reaction of butan-1-ol to butanal, oxidation can be identified by an increase in the oxygen to hydrogen ratio.

Complete the table to show the oxygen to hydrogen ratios in butan-1-ol and butanal.

1

	Oxygen to hydrogen ratio
Butan-1-ol	
Butanal	

[Turn over



* X 8 1 3 7 6 0 1 1 3 *